# Getting help: chatbots and coding agents



# You do not have to memorise this

- Today's commands have dozens of options - nobody remembers them all
- Say what you want in plain words, with your file names
- The chatbot writes the command and explains every option
- Paste an error message the same way

**macs3 callpeak --help: 43 options**



**your question, in plain words**

**the command of step 1.6 - and what each option does**

I have a bedGraph file tracks/ChIP_vs_input_10kb.bedGraph with log2(ChIP/input) in 10 kb bins from bamCompare. I want to find CENH3 domains: regions where the value is at least 3, at least 20 kb long, and gaps of up to 50 kb inside a domain should be joined. Which macs3 command does this? Keep the answer short.

Recalling the bdgbroadcall options from memory ›

bash

```
macs3 bdgpeakcall \
    -i tracks/ChIP_vs_input_10kb.bedGraph \
    -c 3 \
    -l 20000 \
    -g 50000 \
    --no-trackline \
    -o CENH3_domains.narrowPeak
```

What each option does:

- `-c 3` : the cutoff. A bin is "in" when its value is ⩾ 3, so it works directly on your log2 ratios.
- `-g 50000` : the maximum gap. Above-cutoff runs separated by ⩽ 50 kb are merged into one region.
- `-l 20000` : the minimum length. It is applied after merging, so a domain is kept if the merged region is ⩾ 20 kb. With 10 kb bins, that means at least two bins spanning the region.

claude.ai, 2026-11-08 - any chatbot gives a similar answer

own screenshot, claude.ai (Claude Opus 5.5), 2026-11-08

# Three ways to get help

you type more → you type less

## Chat in the browser



ChatGPT · Claude · Gemini · Le Chat
chat.ai.e-infra.cz
(free for Czech academia)

**you give**
a question, an error message,
a pasted output

**it does**
writes the command,
explains the output

## Chat with your files

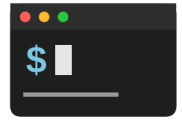


the same chatbots,
with a file attached

**you give**
the fastp report, a flagstat
output, a BED file

**it does**
reads the file, interprets it,
suggests the next step

## Agent in your folder




terminal: Codex · Claude Code
Gemini CLI · opencode
editor: Antigravity · Cursor
VS Code + Copilot

**you give**
the goal - a sentence
or a whole page

**it does**
reads your files, runs commands,
reads output, fixes, tries again

| | Chat in the browser | Chat with your files | Agent in your folder |
|---|---|---|---|
| **who types the command** | you | you | the agent |
| **who checks the result** | you | you | you |

**the checking never moves**

tool names from the vendors' documentation, 2026-11

# The agent types, you think

**1 you**

The file CEN6_genes.gff3 in this folder is the gene annotation of pea chromosome 6. How many genes does it contain, how many are on each strand, and what is the median gene length? Do not modify any files.

**2 agent runs**

```
$ head -20 CEN6_genes.gff3
OX352004  Mikado_loci  gene  2906  6627  .  -  .  ID=…
$ wc -l CEN6_genes.gff3
35887 CEN6_genes.gff3
```

**3 agent runs**

```
$ awk -F'\t' '$3=="gene" {count++; strands[$7]++; …' \
    CEN6_genes.gff3
Total genes: 2028
+ strand: 1052
- strand: 976
```

**4 agent runs**

```
$ awk -F'\t' '$3=="gene" {print $5-$4+1}' CEN6_genes.gff3
    | sort -n | awk '{a[NR]=$1} END {… a[NR/2] …}'
Median gene length: 1463
```

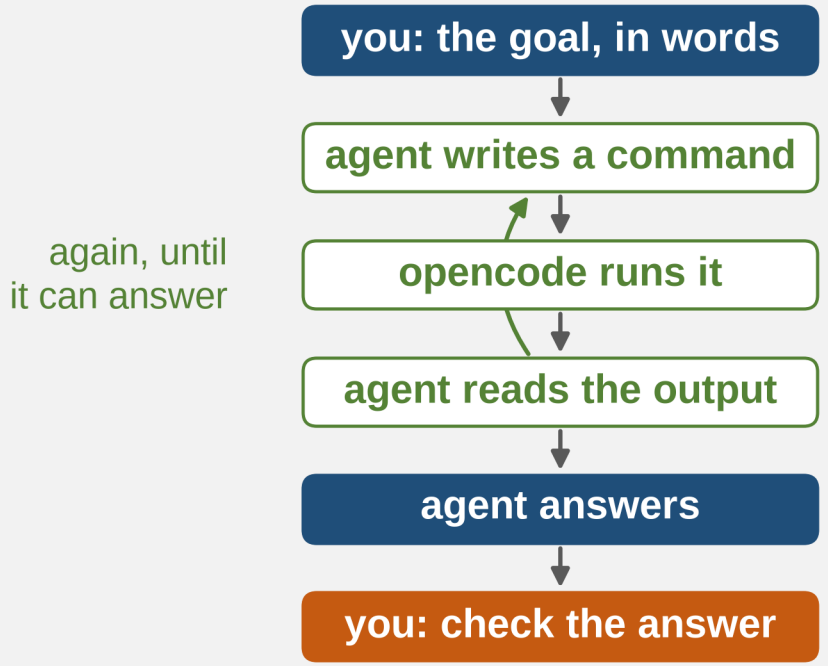
**5 agent checks**

```
$ awk -F'\t' '$3=="gene"' CEN6_genes.gff3 | wc -l; …
2028
1462
1464
```

**6 agent answers**

**2,028 genes: 1,052 on the + strand, 976 on the - strand; median gene length 1,463 bp.**
“Only rows with feature type gene were counted …”

## the loop




**still yours:**
Is “gene” the right feature?
The median of genes - or of transcripts?

opencode 2.2.1 + GLM 5.3 (e-INFRA CZ)
recorded 2026-11-08, 5 shell commands,
shortened where marked …

# What stays yours

- **The question** - which files, which region, what counts as a gene
- **The meaning** - MAPQ 0, ChIP against input, a domain of hundreds of kb
- **The check** - it answers with confidence even when it is wrong; so did MACS3
- **The record** - keep the commands it ran: that is your methods section

